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CASE OF MONTH APRIL 2012

A 52-YEAR-OLD MALE WITH VISUAL CHANGES

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CLINICAL HISTORY AND RADIOLOGY

The patient was a 52 year old male with a history of Cushing's disease, initially diagnosed at the age of 18. At that time, his symptoms included increased weight, purple abdominal striae, diabetes mellitus, and polydipsia. He underwent a resection of a pituitary microadenoma at another hospital. However, due to persistently high cortisol levels following the surgery, he later underwent a bilateral adrenalectomy. Following the operation, he began hydrocortisone therapy, lost weight, and was able to maintain glucose control without diabetes medications. At the age of 32 his diabetes recurred, but he was able to maintain good glucose control up until age 51 at which time his HbA1C was greater than 7. He also had visual problems with significant reduction in right peripheral vision and moderate impairment in left. Testing revealed a greatly elevated serum ACTH at 5,082 pg/mL (normal is 9–46 pg/mL), but serum cortisol within normal limits 4 ug/dL. MRI scans found a large, contrast-enhancing mass in the pituitary fossa with extrasellar extension into the right cavernous sinus and displacement of the parasellar carotid posteriorly (Figs 1a and 1b). The patient underwent endoscopic endonasal approach for debulking of the mass.

PATHOLOGY

Sections showed a cellular tumor (Figs 2 and 3a). Reticulin stains showed variation of reticulin within the tumor (Figs 3b and 3c). Immunostaining for synaptophysin (Fig 3d) ACTH (Fig 4) and cytokeratin (Cam 5.2, Figure 5) were positive in the tumor. Ki67 showed increased proliferation compared to most pituitary adenomas. **What is the diagnosis?**

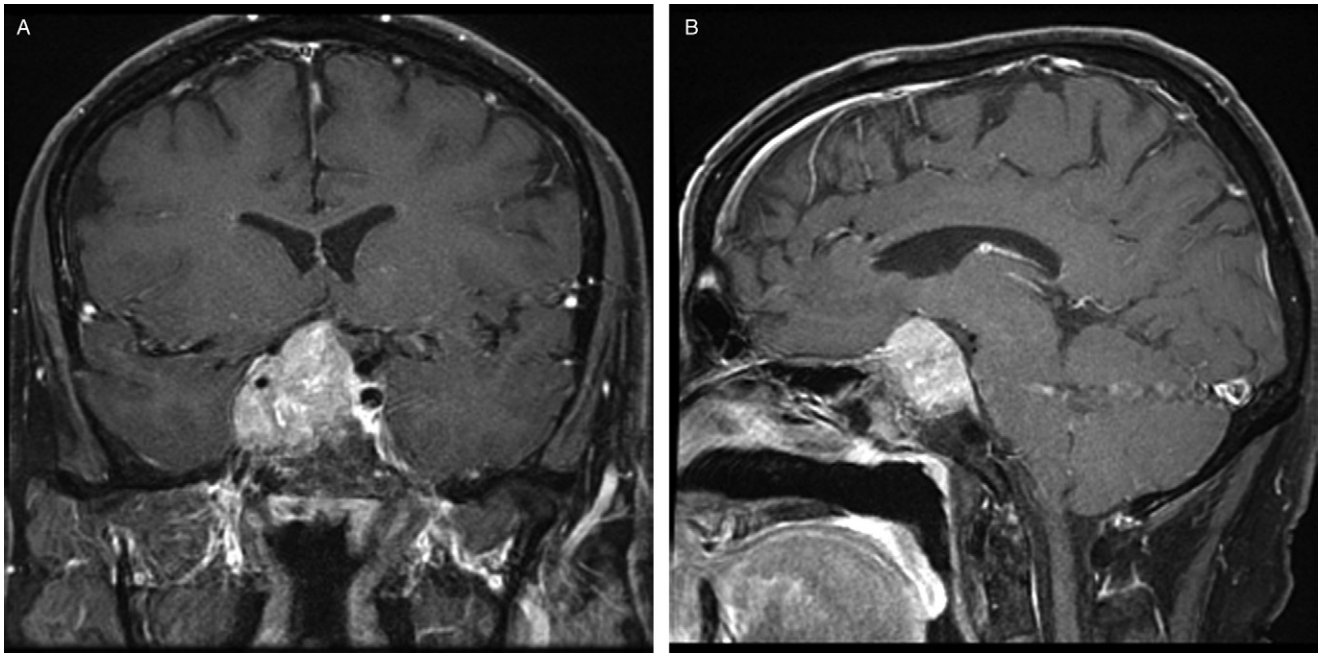


Figure 1.

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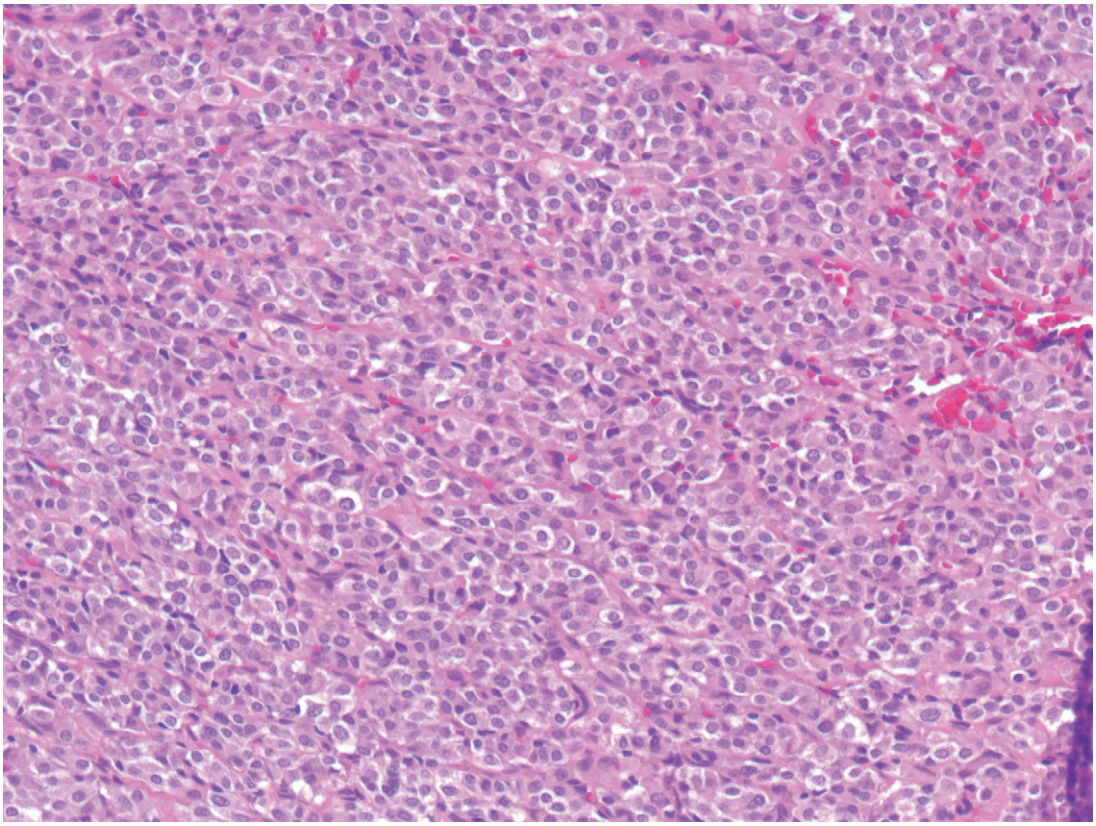


Figure 2.

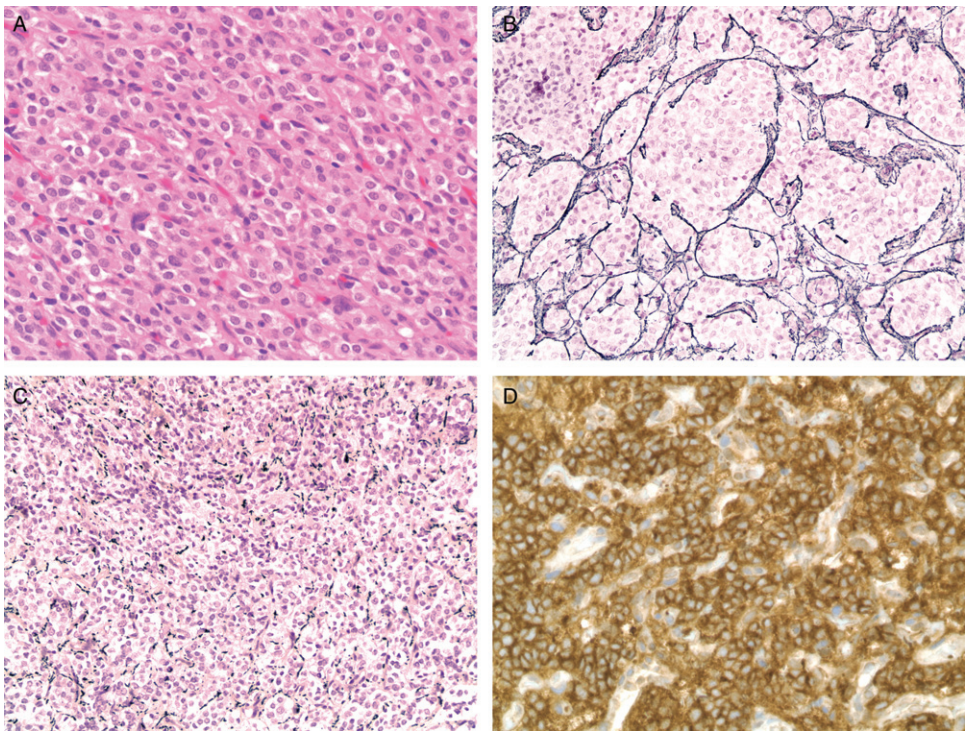


Figure 3.

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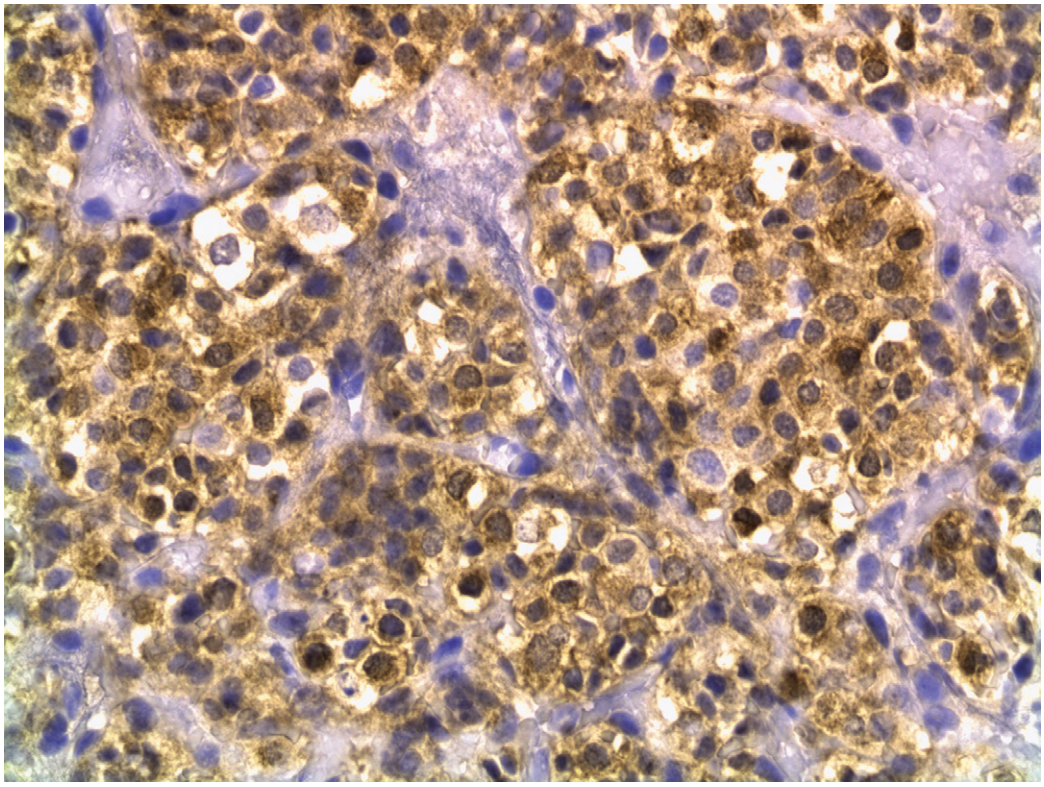


Figure 4.

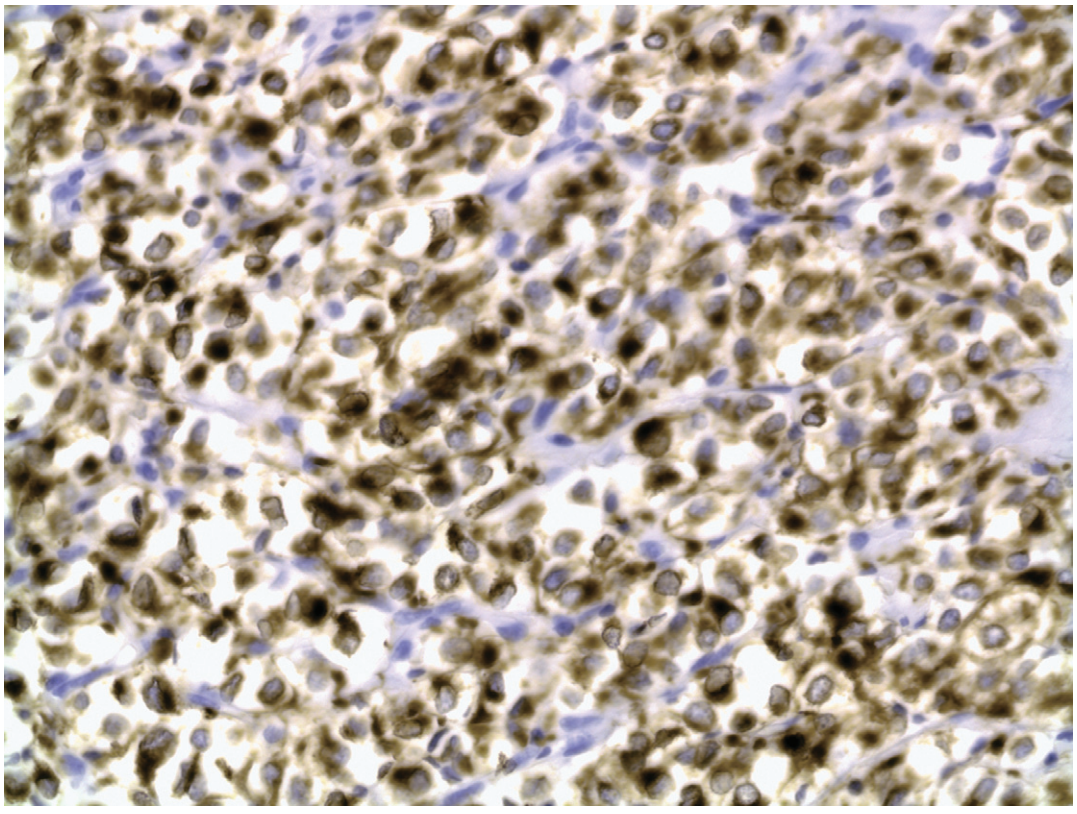


Figure 5.

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DIAGNOSIS

Nelson's Syndrome

DISCUSSION

Cushing's Disease is caused by an adrenocorticotropin hormone (ACTH)-secreting pituitary adenoma. Refractory Cushing's Disease may be treated by bilateral adrenalectomy, which is a definitive treatment, but 20–30% of cases will develop Nelson's syndrome in which the adenoma continues to grow and secrete ACTH (1, 3). Hyperpigmentation is often seen and when the adenoma grows large enough there may be visual field defects or cranial nerve symptoms.

Crooke's hyaline change was first described in 1935 in the normal anterior pituitary surrounding an ACTH-secreting adenoma. The non-neoplastic corticotrophs have increased eosinophilic cytoplasm filled with cytokeratin while the adenoma itself does not. The cause of Crooke's hyaline change is uncertain, but it is related to increased glucocorticoid or cortisol levels. Crooke's cells are found in pituitaries of Cushing's Disease, but not in the pituitary surrounding the adenomas of Nelson's syndrome (4).

In very rare cases the ACTH-secreting adenoma is itself composed of cells with Crooke's hyaline change (Fig 6a). Because of the perinuclear accumulation of cytokeratin, the ACTH stains show ring-like areas of negative staining (Fig 6b). Although pituitary adenomas are cytokeratin positive for CAM5.2, Crooke's cells (whether in the surrounding pituitary or in the adenoma) will show very strong perinuclear and ring-like accumulations of positive-staining (Fig 6c). These tumors are called Crooke's Cell Adenomas and they are an aggressive variant of ACTH-secreting adenomas. In 2003, 36 cases of Crooke's Cell Adenomas were examined and compared to the 7 cases previously published (2). None of these adenomas were in cases of Nelson's Syndrome.

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Acknowledgment: Figures 6a,b,c in the discussion are courtesy of Dr. Arie Perry.

ABSTRACT

A 52 year-old male with a history of Cushing's Disease at age 18 and bilateral adrenalectomy at age 23 presented with visual changes. An MRI scan showed a pituitary macroadenoma (Nelson's syndrome). Other than the development of diabetes mellitus at age 32, his disease was stable until presentation. Serum studies show markedly elevated ACTH levels, but he had no hyperpigmentation. The tumor was removed by endoscopic surgery. Microscopic examination showed a pituitary adenoma with strong immunostaining for ACTH. In addition, the tumor cells showed Crooke's hyaline change and stained strongly for cytokeratin (Crooke's Cell Adenoma). Normal pituitary was not present. Crooke's cell adenomas are extremely rare and have not been previously reported in Nelson's Syndrome.

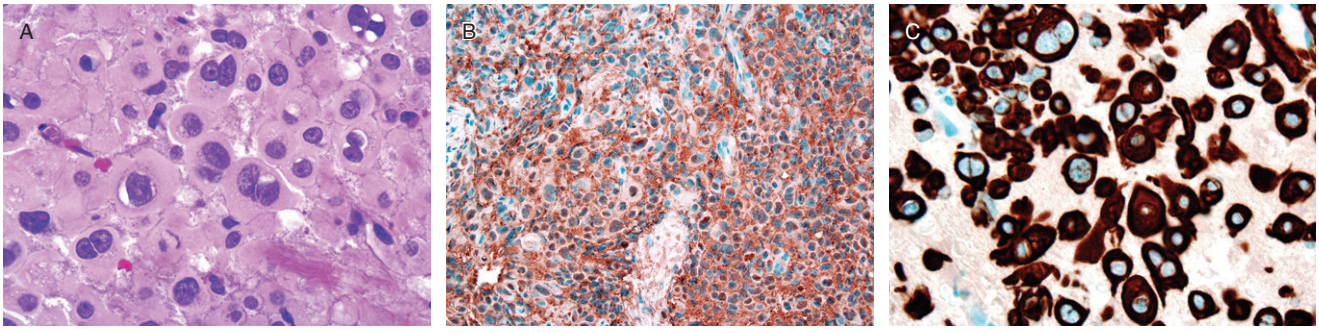


Figure 6.

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